

Temporal Gap Analysis

GBIF Gap Analysis - Temporal Patterns & Data Recency

gbifgaps

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Overview

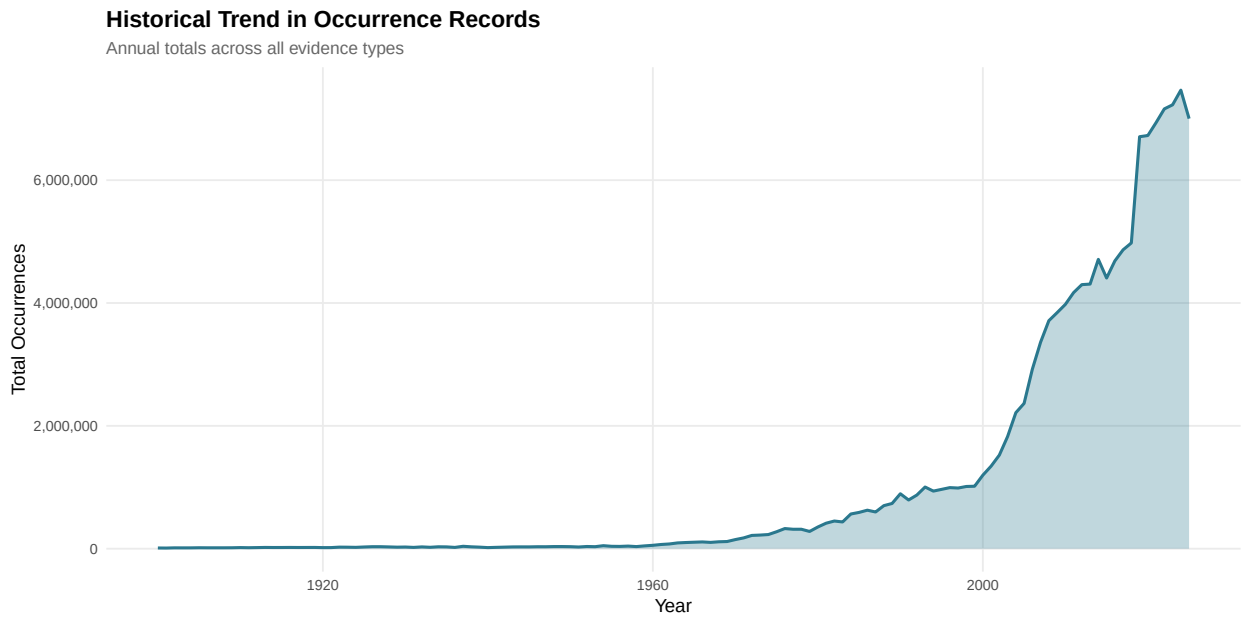
This report analyzes **temporal gaps** in the national biodiversity occurrence data, examining historical trends, seasonal patterns, data recency, and how data mobilization has changed over time for different taxonomic groups.

Key questions addressed:

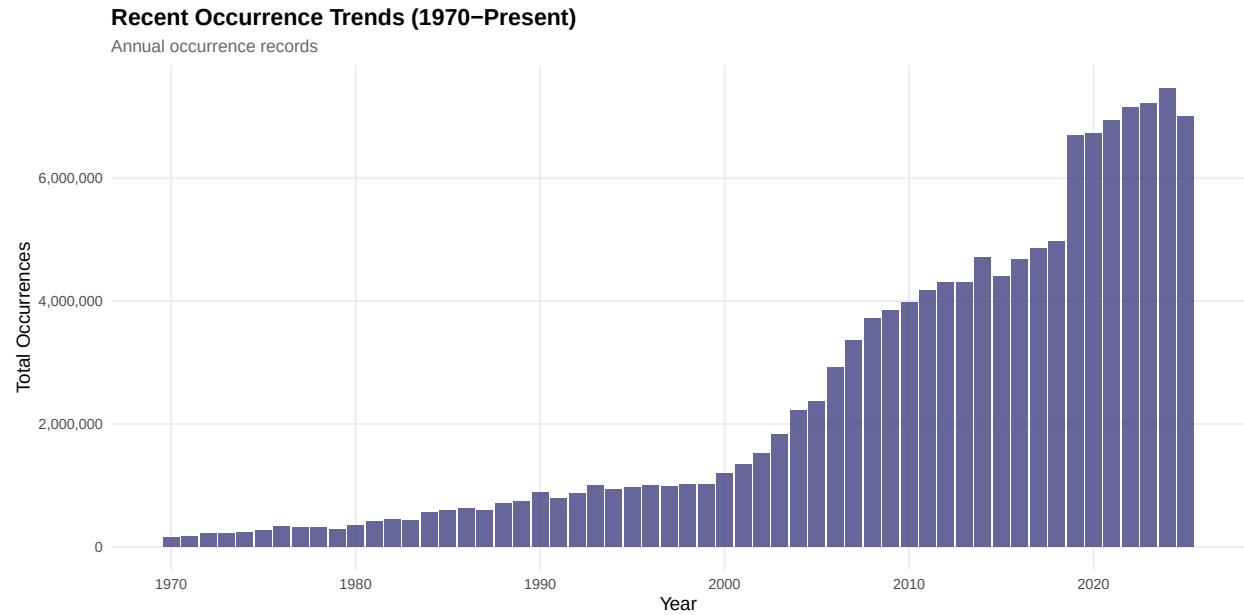
- What are the long-term trends in biodiversity data collection?
- Are there seasonal biases in sampling?
- How recent is the data in different geographic areas?
- How has data mobilization changed for different taxonomic orders over 5-year periods?

1. Historical Trends

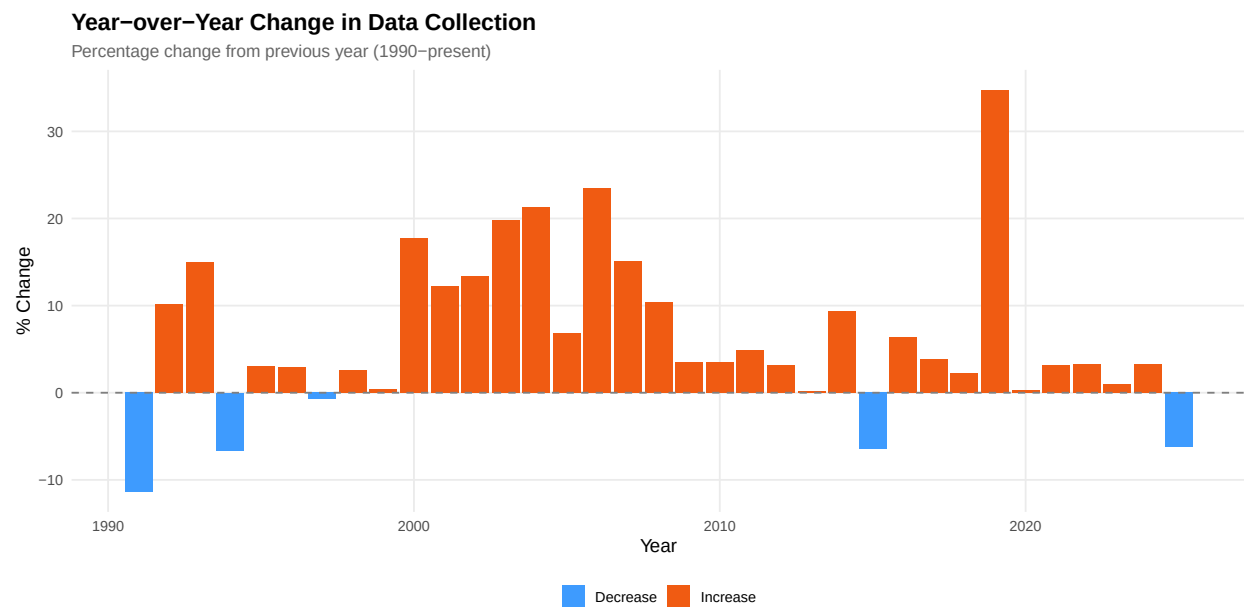
1.1 Annual Occurrence Trends



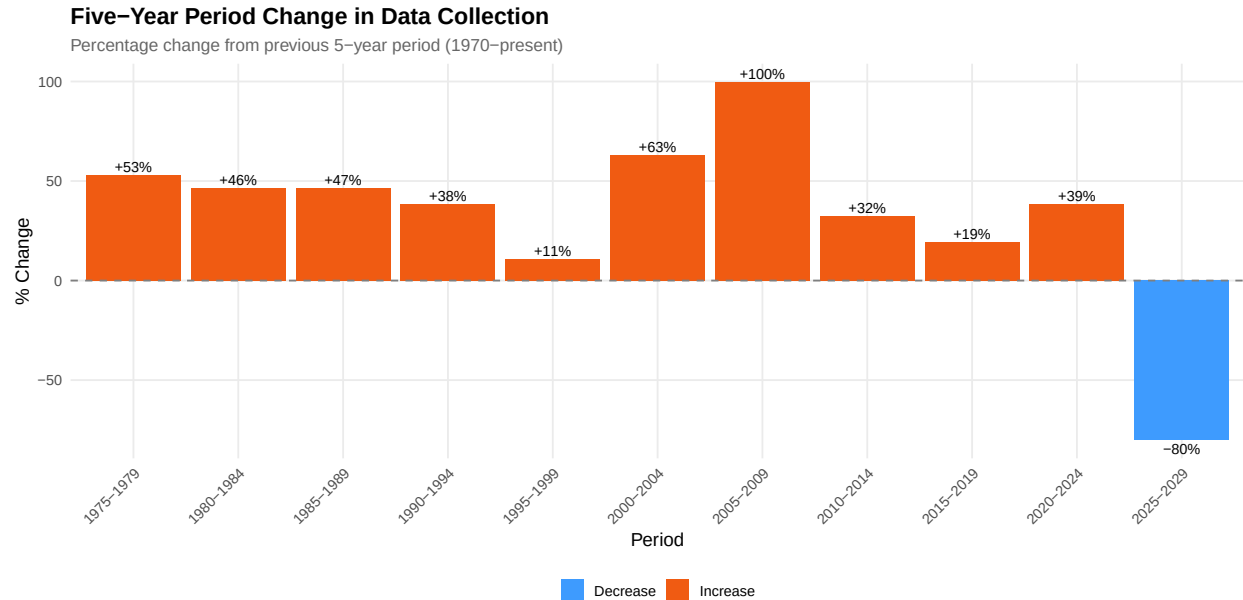
1.2 Recent Decades (1970-Present)



1.3 Year-over-Year Change

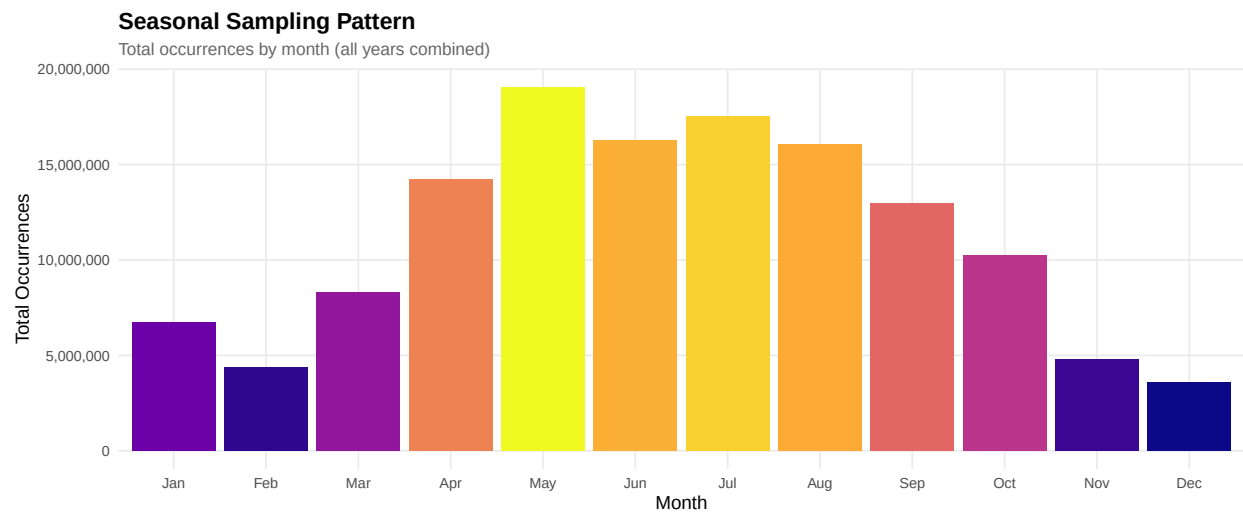


1.4 Five-Year Period Change

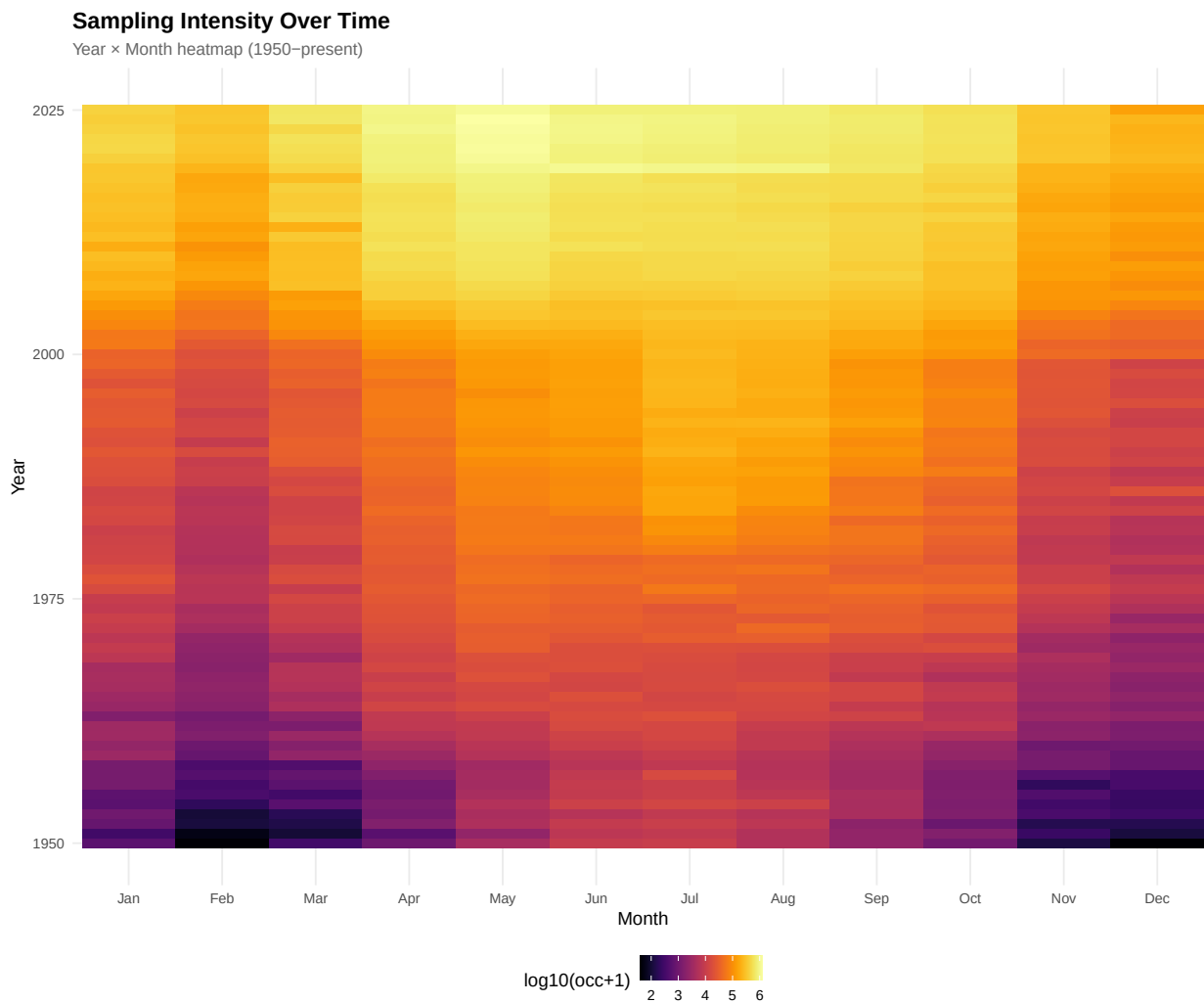


2. Seasonal Patterns

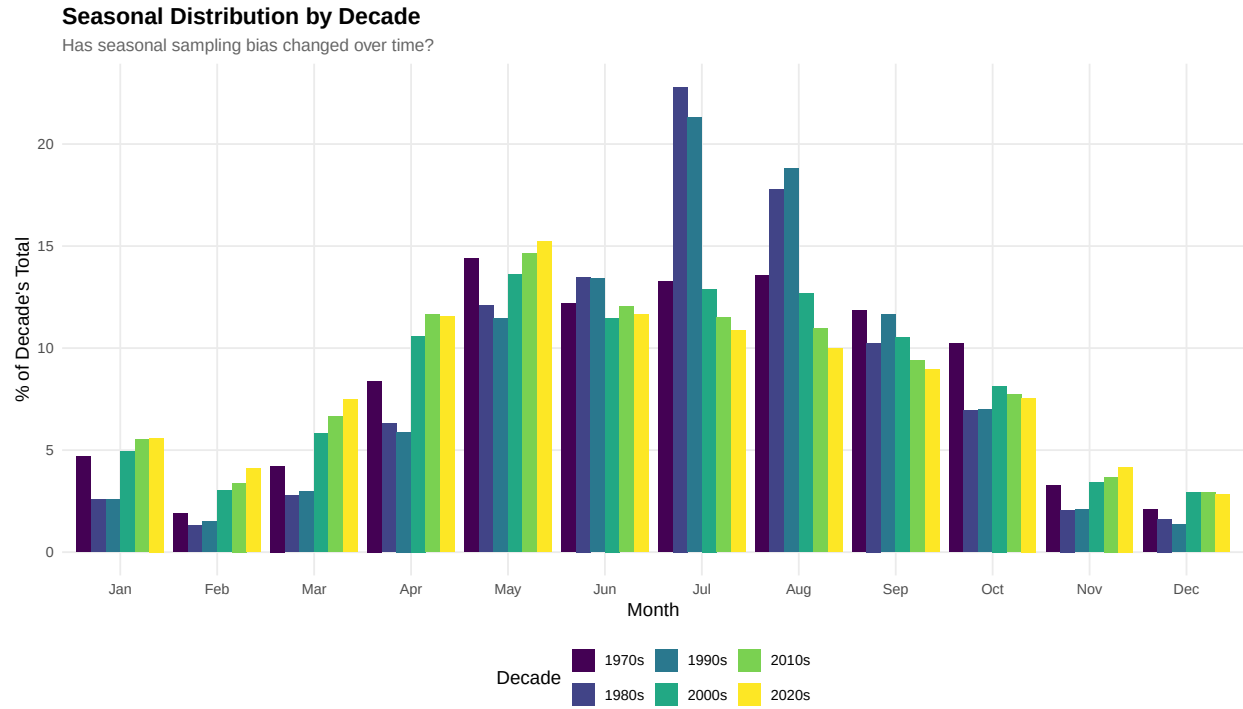
2.1 Monthly Distribution



2.2 Year × Month Heatmap



2.3 Seasonal Bias by Decade



3. Data Recency

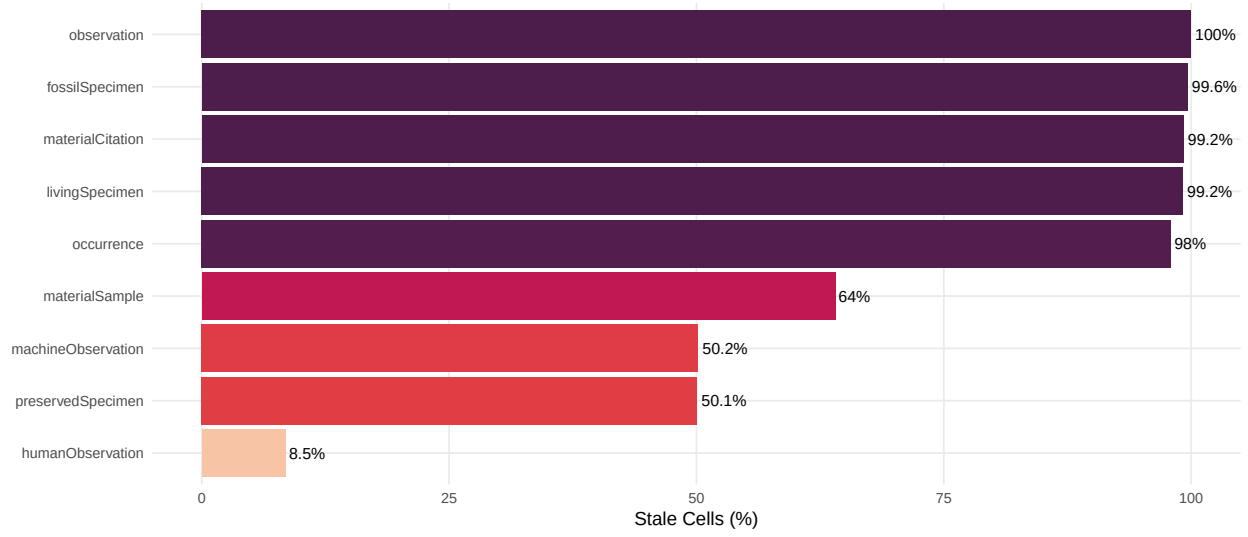
3.1 Staleness by Basis of Record

Table 1: Data Staleness by Evidence Type (10km Grid)

basisofrecord	median_staleness_months	pct_stale_5y
humanObservation	3	8.5
all	4	23.5
preservedSpecimen	62	50.1
machineObservation	63	50.2
materialSample	86	64.0
occurrence	1075	98.0
livingSpecimen	269	99.2
materialCitation	199	99.2
fossilSpecimen	1148	99.6
observation	403	100.0

Data Staleness by Evidence Type

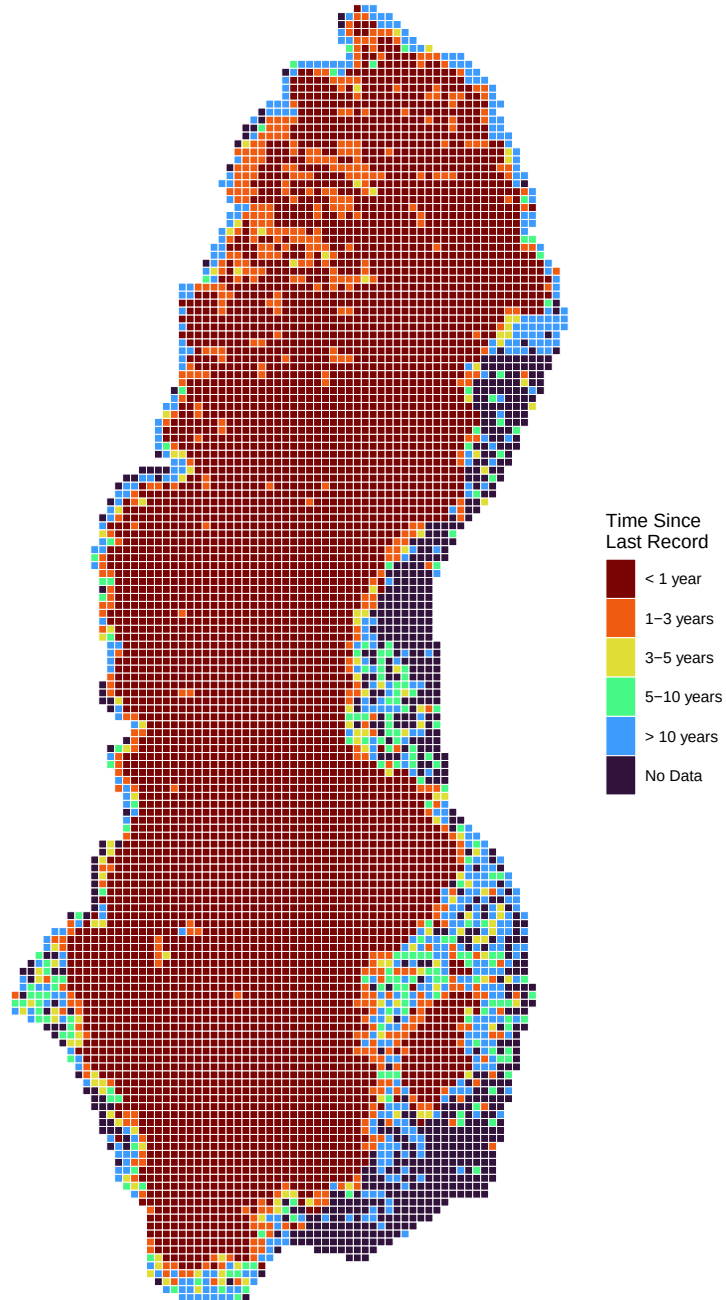
Percentage of cells not sampled in 5+ years



3.2 Geographic Distribution of Staleness

Geographic Distribution of Data Recency

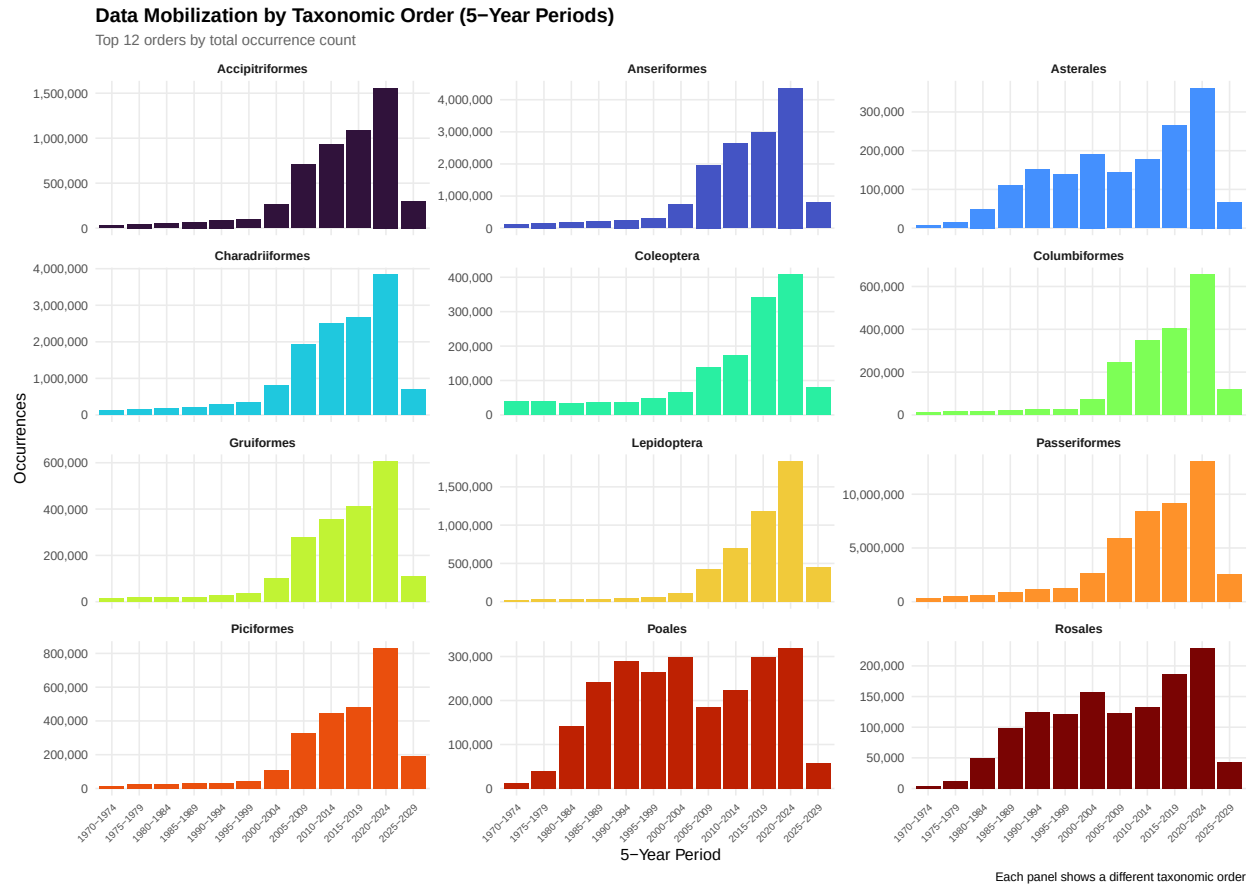
Time since last occurrence record per cell



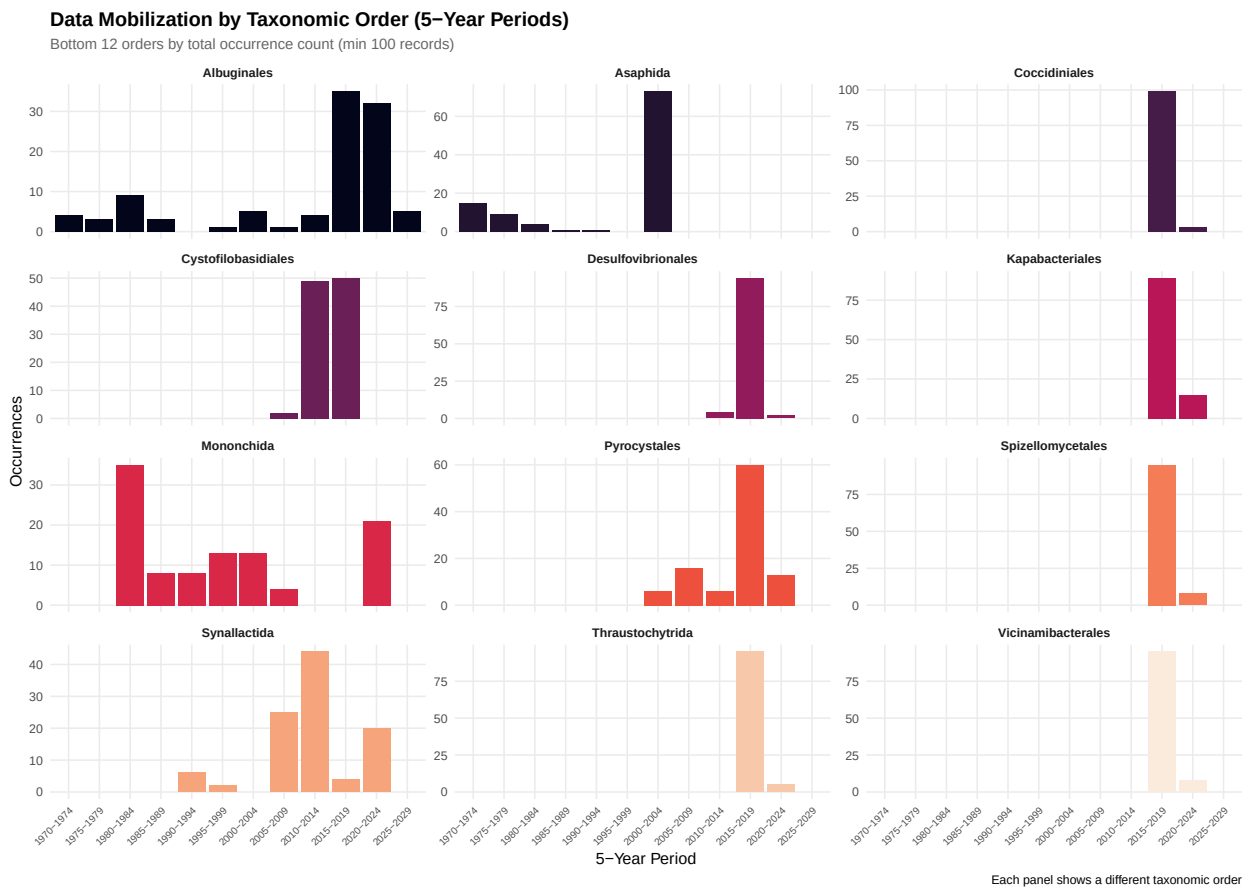
4. Data Mobilization by Taxonomic Order (5-Year Periods)

This section examines how data mobilization has changed over time for different taxonomic orders, using 5-year time bins.

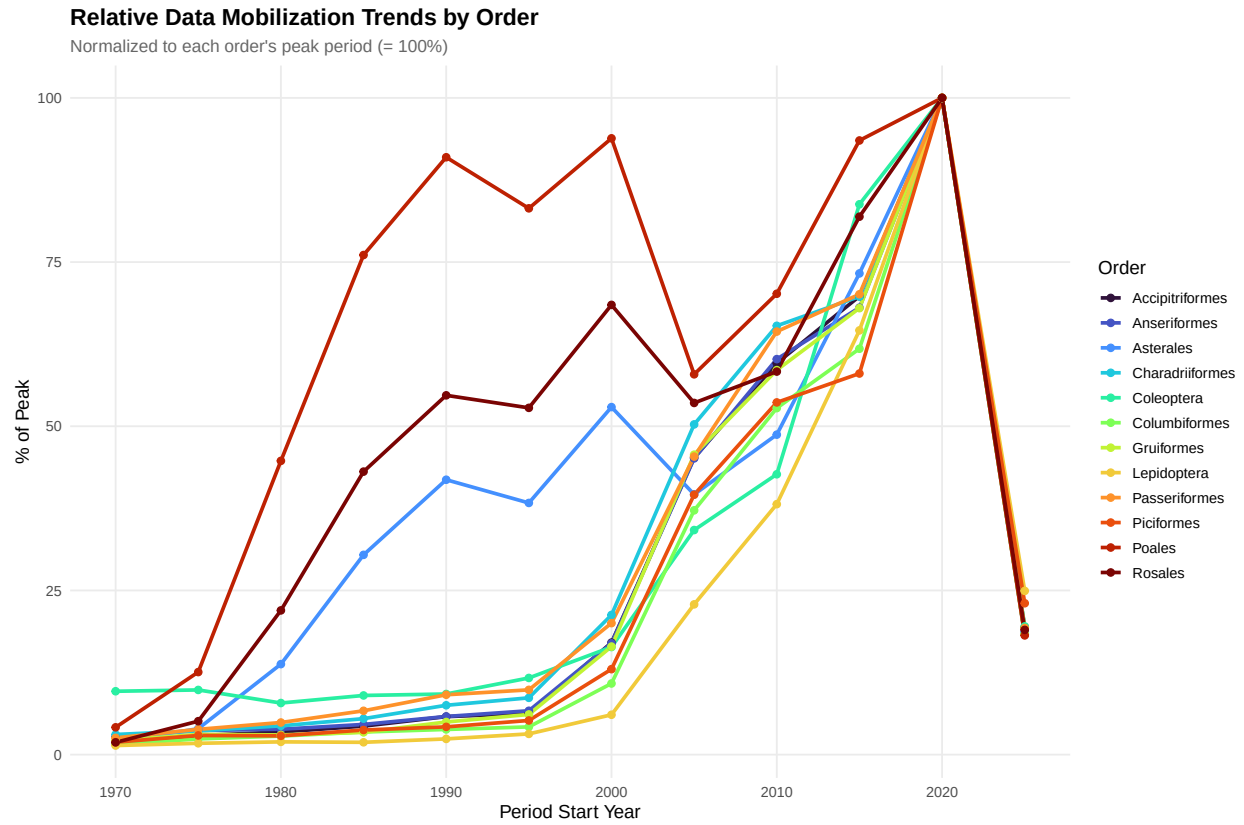
4.1 Data Mobilization Trends by Order (Top 12)



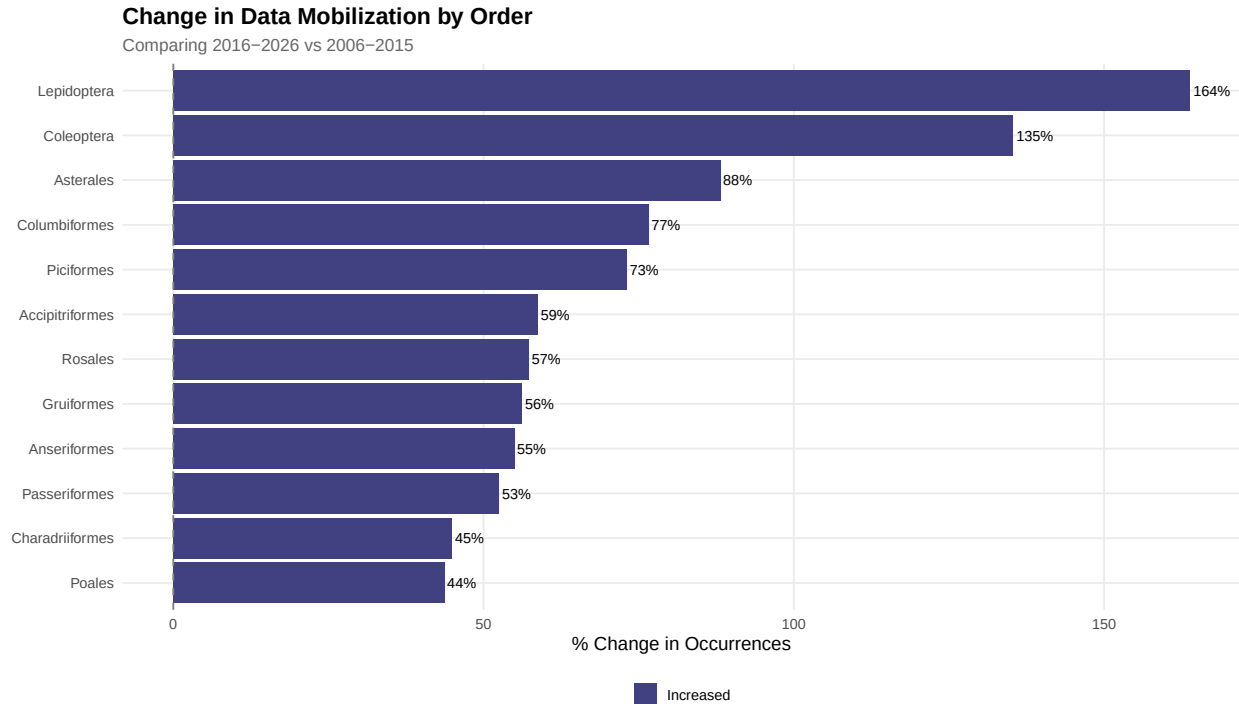
4.2 Data Mobilization Trends by Order (Bottom 12)



4.3 Comparative Mobilization Trends



4.4 Change in Mobilization: Recent vs Historical



4.5 5-Year Mobilization Summary Table

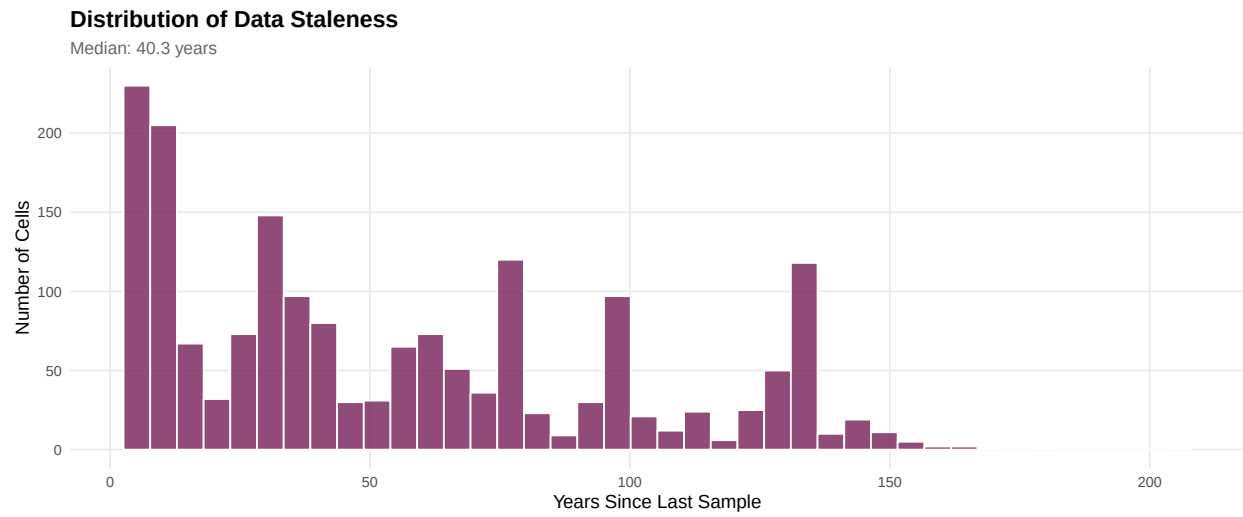
Table 2: Occurrences by Order and 5-Year Period

order	1970- 1974	1975- 1979	1980- 1984	1985- 1989	1990- 1994	1995- 1999	2000- 2004	2005- 2009	2010- 2014	2015- 2019	2020- 2024	2025- 2029	Total
Passeriformes	383,080	502,246	638,038	870,905	1,189,978	1,287,090	1,613,481	1,923,181	2,407,740	2,148,673	2,051,323	2,316,596	16,482,412
Anseriformes	19,543	160,141	168,973	200,900	253,733	291,803	374,412	1,971,172	2,631,508	2,973,991	3,369,588	3,160,146	14,682,918
Charadriiformes	7,707	138,931	168,172	210,166	288,331	331,800	816,152	1,929,562	2,505,642	2,676,675	2,837,396	3,044,404	13,724,938
Accipitriformes	26,604	45,175	53,788	67,660	90,217	97,823	265,959	705,173	929,481	1,088,334	1,559,323	2,302,318	5,232,157
Lepidoptera	25,358	31,612	35,594	34,370	43,848	57,814	111,181	419,105	698,622	1,183,332	2,831,849	4,566,747	9,929,432
Piciformes	15,733	24,234	23,773	31,144	35,122	43,202	108,109	328,823	445,529	482,096	830,748	1,482,559	2,559,995
Poales	13,259	40,095	142,667	242,637	290,115	265,305	299,301	184,695	223,833	298,266	318,942	57,897	2,377,012
Gruiformes	3,018	18,626	18,176	20,601	30,095	37,044	99,641	276,615	354,605	411,735	605,649	111,190	1,996,995
Columbiformes	10,686	15,687	18,518	22,553	25,126	27,654	70,986	243,743	334,738	404,983	655,248	119,925	1,960,247
Asterales	6,937	14,312	49,913	110,183	151,618	138,817	191,689	143,503	176,491	265,420	362,224	67,648	1,678,755
Coleoptera	39,327	40,140	32,025	36,717	37,619	47,574	66,776	139,364	173,976	341,519	407,543	79,572	1,442,152
Rosales	4,272	11,657	50,270	98,647	125,233	120,870	156,760	122,582	133,517	187,513	228,906	43,479	1,283,706

5. Stale Cells Analysis

5.1 Distribution of Staleness

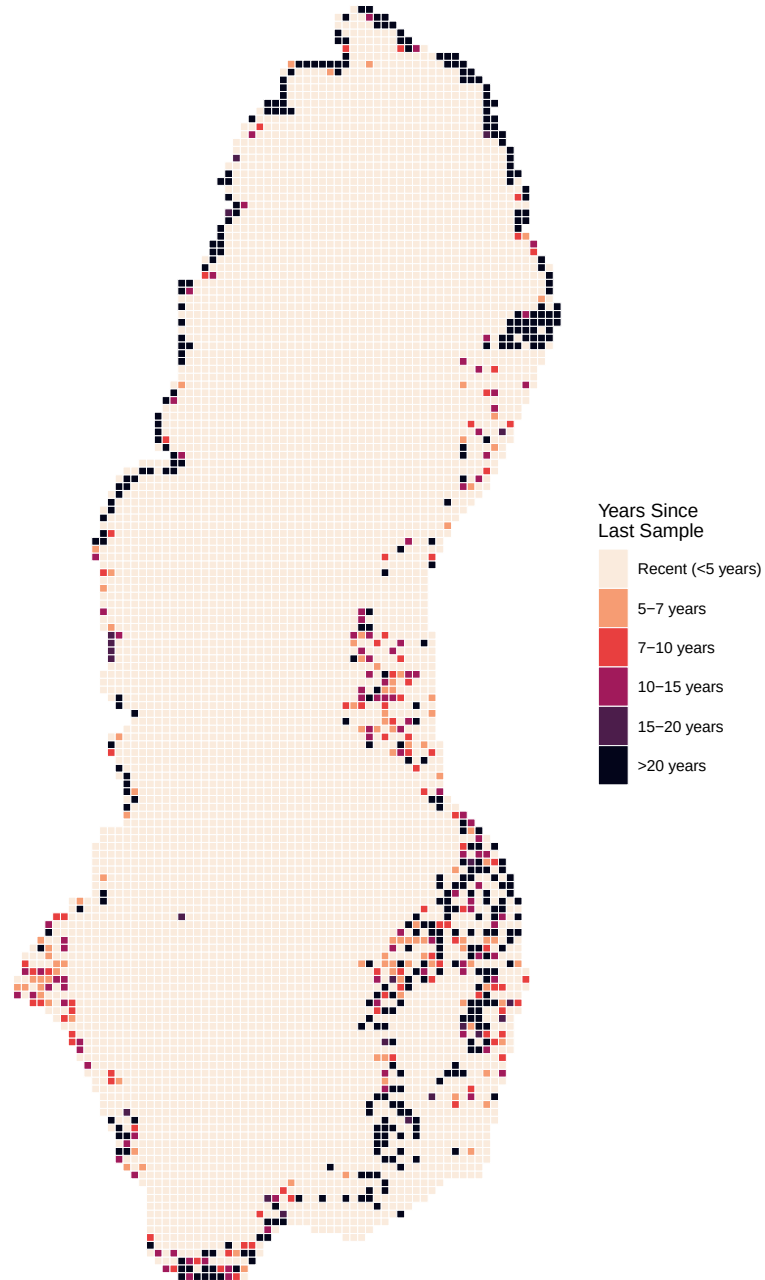
** 1,805 cells** have not been sampled in 5+ years.



5.2 Most Stale Cells

Priority Stale Cells by Staleness

1,805 cells not sampled in 5+ years



Darker colors indicate longer time since last observation

Table 3: Top 20 Most Stale Cells

grid	eeacellcode	last_ym	years_stale	total_occurrences
grid10km	10kmE516N475	1820-08-01	205.4	1
grid10km	10kmE413N466	1844-12-01	181.1	1
grid10km	10kmE454N482	1856-08-01	169.4	1
grid10km	10kmE491N501	1859-07-01	166.5	1
grid10km	10kmE495N501	1859-09-01	166.3	2
grid10km	10kmE499N481	1864-08-01	161.4	2
grid10km	10kmE437N406	1867-04-01	158.8	1
grid10km	10kmE511N450	1869-07-01	156.5	1
grid10km	10kmE490N427	1869-11-01	156.2	1
grid10km	10kmE527N433	1870-06-01	155.6	1
grid10km	10kmE511N406	1871-08-01	154.4	1
grid10km	10kmE391N433	1873-06-01	152.6	1
grid10km	10kmE514N484	1874-09-01	151.3	1
grid10km	10kmE411N407	1875-07-01	150.5	1
grid10km	10kmE504N480	1875-08-01	150.4	1
grid10km	10kmE530N406	1876-02-01	149.9	1
grid10km	10kmE543N445	1876-05-01	149.7	1
grid10km	10kmE456N493	1876-07-01	149.5	1
grid10km	10kmE534N437	1877-12-01	148.1	1
grid10km	10kmE406N453	1877-12-01	148.1	1

6. Key Findings

Temporal Gap Summary

Time Range:

- Data spans: 1900 - 2025
- Peak year: 2024 (7,465,457 occurrences)

Seasonal Patterns:

- Peak month: May
- Summer (Jun-Aug) accounts for: 37.2 % of records

Data Recency:

- Median staleness: 4 months
- Cells stale >5 years: 23.5 %

Mobilization Trends by Order:

- Orders with increased mobilization: 12
- Orders with decreased mobilization: 0

- Largest increase: Lepidoptera (+ 164 %)

Recommendations:

1. Re-survey cells with >5 years since last observation
2. Address seasonal sampling bias with winter surveys
3. Maintain momentum for orders showing increased mobilization
4. Investigate causes for orders showing decreased mobilization

Session Info

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## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.2
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices datasets  utils      methods    base
##
## other attached packages:
## [1] tibble_3.3.1      kableExtra_1.4.0  knitr_1.50        patchwork_1.3.2
## [5] forcats_1.0.1     fstcore_0.10.0    shinyjs_2.1.1     cli_3.6.5
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## [13] here_1.0.2        lubridate_1.9.4   glue_1.8.0        viridis_0.6.5
## [17] viridisLite_0.4.2 scales_1.4.0       DT_0.34.0         sf_1.0-24
## [21] leaflet_2.2.3     plotly_4.12.0     ggplot2_4.0.1     tidyr_1.3.2
## [25] dplyr_1.1.4       shinyWidgets_0.9.0 shiny_1.12.1
##
## loaded via a namespace (and not attached):
## [1] DBI_1.2.3          gridExtra_2.3      rlang_1.1.7
## [4] magrittr_2.0.4     otl_0.2.0          e1071_1.7-17
## [7] compiler_4.5.2     systemfonts_1.3.1  callr_3.7.6
## [10] vctr_0.7.0         crayon_1.5.3       pkgconfig_2.0.3
## [13] fastmap_1.2.0      backports_1.5.0    labeling_0.4.3
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## [19] rmarkdown_2.30     tzdb_0.5.0         ps_1.9.1
## [22] tinytex_0.58       bit_4.6.0          xfun_0.56
## [25] cachem_1.1.0       jsonlite_2.0.0     later_1.4.5
## [28] tarchetypes_0.13.2 terra_1.8-93        parallel_4.5.2
## [31] prettyunits_1.2.0  R6_2.6.1           bslib_0.9.0
## [34] stringi_1.8.7      RColorBrewer_1.1-3 jquerylib_0.1.4
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## [46] codetools_0.2-20	processx_3.8.6	withr_3.0.2
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## [52] proxy_0.4-29	xml2_1.5.2	pillar_1.11.1
## [55] BiocManager_1.30.27	KernSmooth_2.23-26	renv_1.1.6
## [58] generics_0.1.4	vroom_1.6.7	rprojroot_2.1.1
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## [64] class_7.3-23	lazyeval_0.2.2	tools_4.5.2
## [67] fs_1.6.6	grid_4.5.2	crosstalk_1.2.2
## [70] textshaping_1.0.4	fst_0.9.8	svglite_2.2.2
## [73] gt_1.3.0	gtable_0.3.6	sass_0.4.10
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## [79] farver_2.1.2	memoise_2.0.1	htmltools_0.5.8.1
## [82] lifecycle_1.0.4	httr_1.4.7	secretbase_1.1.1
## [85] mime_0.13	bit64_4.6.0-1	